

Caffeine for treatment of Parkinson disease

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Objective: Epidemiologic studies consistently link caffeine, a nonselective adenosine antagonist, to lower risk of Parkinson disease (PD). However, the symptomatic effects of caffeine in PD have not been adequately evaluated. Methods: We conducted a 6-week randomized controlled trial of caffeine in PD to assess effects upon daytime somnolence, motor severity, and other nonmotor features. Patients with PD with daytime somnolence (Epworth >10) were given caffeine 100 mg twice daily ×3 weeks, then 200 mg twice daily ×3 weeks, or matching placebo. The primary outcome was the Epworth Sleepiness Scale score. Secondary outcomes included motor severity, sleep markers, fatigue, depression, and quality of life. Effects of caffeine were analyzed with Bayesian hierarchical models, adjusting for study site, baseline scores, age, and sex. Results: Of 61 patients, 31 were randomized to placebo and 30 to caffeine. On the primary intention-to-treat analysis, caffeine resulted in a nonsignificant reduction in Epworth Sleepiness Scale score (−1.71 points; 95% confidence interval [CI] −3.57, 0.13). However, somnolence improved on the Clinical Global Impression of Change (+0.64; 0.16, 1.13, intention-to-treat), with significant reduction in Epworth Sleepiness Scale score on per-protocol analysis (−1.97; −3.87, −0.05). Caffeine reduced the total Unified Parkinson's Disease Rating Scale score (−4.69 points; −7.7, −1.6) and the objective motor component (−3.15 points; −5.50, −0.83). Other than modest improvement in global health measures, there were no changes in quality of life, depression, or sleep quality. Adverse events were comparable in caffeine and placebo groups. Conclusions: Caffeine provided only equivocal borderline improvement in excessive somnolence in PD, but improved objective motor measures. These potential motor benefits suggest that a larger long-term trial of caffeine is warranted. Classification of evidence: This study provides Class I evidence that caffeine, up to 200 mg BID for 6 weeks, had no significant benefit on excessive daytime sleepiness in patients with PD.